

PAWCONNECT: A QR-BASED PROFILE TRACKING SYSTEM FOR LOST CATS AND DOGS

Local Literature

1. Cat's Paw: Tracking the Home Range of Domestic Cats in Mount Makiling Forest Reserve, Tenorio et al. (2024). This journal Discusses the roaming patterns along with the predation behaviors of domestic cats in the Mount Makiling Forest Reserve, Laguna, Philippines. The journal shows that domestic cats, whether owned or not, can significantly impact local wildlife, preying on native species and potentially disrupting the ecosystem. The authors emphasized the need for responsible pet ownership, this includes registration, sterilization, and limiting outdoor access for cats, especially in biodiversity rich areas. These recommendations align with the need to improve responsible pet management practices, which supports initiatives for better pet welfare and adoption systems.
2. Dog Ecology and Demographics in Several Areas in the Philippines and Its Application to Anti-Rabies Vaccination Programs, Chaudhari et al. (2022). This article shows an investigation regarding dog ecology and demographics across the Philippines, it emphasizes the importance of accurate population estimates when it comes to planning effective anti-rabies vaccination programs. The article demonstrated the utility of mobile applications for dog demographic surveys in urban and rural areas, showing how technology can facilitate efficient data gathering analysis. Similarly, implementing QR code technology in collars could produce innovative solutions for tracking and managing pets while ensuring rapid access to owner and vaccination information.
3. Philippine Animal Welfare Society (PAWS): Get your pet microchipped and consider providing a pet tag hanging on your pet's collar. If your pet gets lost, this lets good samaritans know who to contact if your pet is found. The microchip is also one way of proving that you own that particular animal.
--- How helpful is a microchip?? Last May 2013, a husky ran out from his home in a posh Makati village and traversed through EDSA until he reached Ortigas. He was reunited with his owner 10 hours later because he was identified to have a microchip - even if his owner did not have any recent photos to show to prove ownership, the owner was able to provide the microchip info that matched the microchip found on the husky when he was scanned. A microchip implant is an identifying integrated circuit placed under the skin of a dog, cat, horse or other animal by a licensed veterinarian. The chip, about the size of a large grain of rice, uses passive RFID (Radio Frequency Identification) technology. Microchips can

be used by vet clinics, rescue groups, kennels, trainers, and registries to identify the animals

4. Solon seeks to address rising cases of animal abandonment: Senator Mark Villar on Thursday called for the passage of a bill that seeks to encourage the adoption of stray and impounded animals, as the rising number of abandoned animals remains highly disproportionate to the support they ought to receive despite projects and shelters that rescue them. In pushing for Senate Bill No. 2257, or the Act Strengthening the Adoption of Stray and Impounded Animals, Villar expressed his concern over the low pet adoption rate in the country, which is less than one percent, amid a large number of homeless animals. According to the Philippine Animal Welfare Society (PAWS), there are around 12 million stray cats and dogs in the Philippines as of 2019, which are either former family pets or those who have reproduced in the streets due to lack of proper shelter.

Local Studies

1. PetPal: Barangay Pet Registration and Monitoring System, Crasicas (2024). This study is about an innovative information system proposed to address the challenges barangays face in managing and monitoring pets. It focuses on promoting responsible pet ownership, ensuring pet's well-being, and enhancing community safety. The focus is on issues like lacking centralized pet data, inefficient record-keeping, pet welfare concerns, unregistered and abandoned pets, limited awareness of pet ownership responsibilities, ineffective regulation enforcement, inaccurate census data, and privacy concerns.
2. Stray Dogs Overpopulation in Municipality of Dauis: Cellular Phone Tracking Technology, Aisa et al. (2021). This study focuses on discussing the negative impacts of stray dogs on the environment and public health, particularly in Dauis, Bohol. It proposes the development of a GPS-based tracking system to monitor stray dogs effectively. The system comes with a prototype collar equipped with a GPS tracker, accelerometer, battery, and SD card, all housed in a 3D-printed case. The collar is designed to record the dogs' locations and store the data on a server for real-time tracking through a mobile device. The study aims to collaborate with a local animal welfare organization, to support their efforts in managing the stray dog population through methods like Catch, Neuter, Vaccine, Release, and Manage (CNVRM). It also aims to enhance understanding of stray dogs' behavior and address overpopulation issues through data collection and tracking.
3. Computerized Pet Registration for Barangay Talon Uno Las Piñas City: According to Agaloos et al. (2022) claims that animals mainly appear as human's best friend nowadays. However, they also can be a threat to society. Large number of cases reported of animal-related accidents, such as car accidents, viral attacks, and animal-induced harm have been reported globally. These incidents often lead to fatalities involving both humans and animals, and the authors cite that such accidents occur without assigning blame to any particular subject. The researcher decided to develop a system that would

make it easy for the staff to retrieve the missing pet files and reports, as well as the owners' details. And by using it, it will help to minimize paper consumption.

4. Detepet: Mobile application for pet tracking: Providing sympathy to those who have lost their pets is a valid feeling. Losing someone's pets is not easy for them. Owners having difficulty finding exactly where their pets are. In this day and age, Technology and software can be a platform between the owner and community. In this case, somehow it solves the problem immediately. Technology plays a vital role in our daily lives nowadays, yet there are still individuals who are experiencing difficulty with the occurrence of smartphones. Everyone must be able to live everyday with technology. Fur parents are having difficulties controlling each vital aspect of their pets. (Aqraldo et. al. , 2021)

Foreign Literature

1. IoT applications for monitoring companion animals: A systematic literature review, Junior (2020) presents the main focus of academic studies in the last five years about what was being investigated in the field of monitoring companion animals. The author also made a few recommendations to further research in order to improve animal welfare.
2. Detepet: A Mobile App for Tracking Lost Pets (2023). This study looks at the problems pet owners face when their pets go missing. Losing a pet often causes stress, and some owners may even give up looking for them. Detepet is a mobile app created to help owners find their pets quickly and easily. Pet owners can register their pets in the app, and a GPS collar is attached to the pet. This collar tracks the pet's location in real time, allowing the owner to see where their pet is at any moment. The goal of the app is to make it easier for owners to keep their pets safe and prevent them from getting lost.
3. PetTag System: An RFID-Based Solution for Cat Monitoring (2023). This study highlights the challenges faced by Malaysian cat owners, including loss, theft, and unforeseen accidents. To address these issues, the PetTag system was developed using a prototype model method. The system integrates RFID technology with an ESP32 module, installed at the cage door, to monitor the entry and exit of cats. Owners can check their cat's status—whether inside or outside the cage—through the system. The electronic components are programmed using Arduino IDE, while MySQL is used to store data.

Foreign Studies

1. Trackable Pet and Livestock Collar, Reddy (2020) discusses the development of a GPS-based collar designed for monitoring pets and livestock. This innovative device incorporates location tracking, environmental monitoring, and data transmission capabilities, providing a practical solution for animal management. The study emphasizes how this technology enables owners to ensure the safety and welfare of their animals, especially in rural and farming settings. The collar features a combination of GPS and GSM technology to transmit real-time location data. Additionally, it includes sensors to monitor environmental parameters like temperature and humidity, crucial for assessing animal health. The authors highlight the significance of this device in

addressing common challenges such as theft prevention, locating lost animals, and monitoring livestock grazing patterns.

2. Stray Dogs Overpopulation in Municipality of Dauis: Cellular Phone Tracking Technology, Aisa et al. (2021). This study focuses on discussing the negative impacts of stray dogs on the environment and public health, particularly in Dauis, Bohol. It proposes the development of a GPS-based tracking system to monitor stray dogs effectively. The system comes with a prototype collar equipped with a GPS tracker, accelerometer, battery, and SD card, all housed in a 3D-printed case. The collar is designed to record the dogs' locations and store the data on a server for real-time tracking through a mobile device. The study aims to collaborate with a local animal welfare organization, to support their efforts in managing the stray dog population through methods like Catch, Neuter, Vaccine, Release, and Manage (CNVRM). It also aims to enhance understanding of stray dogs' behavior and address overpopulation issues through data collection and tracking.
3. According to D. Bhadane, P. Khirude, O. Chavan and A. Lokare (2023), By raising awareness and enhancing the adoption process a web-based platform for animal shelters can be extremely helpful in resolving the difficulties in pet adoption in India.
4. A study performed by M. Blaer (2022) about "Animal rescue tourism: digital technology-enhanced approaches to support voluntourist engagement, animal welfare and rights." This study builds a case study using a multi-method approach to investigate how these organizations may include volunteer travellers in promoting animal care and rights. According to research, Animal Aid uses digitally enhanced methods, mainly social media and information and communication technologies, to increase public knowledge of and participation in animal welfare and rights issues, including through tourism.

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